

Explainable Edge-Aware Graph Attention and BiLSTM Framework for Spatiotemporal Crash Risk Forecasting in Mixed Traffic: An Empirical Study on Dhaka Traffic Video

Abstract—Mixed, non-lane-based traffic in cities such as Dhaka makes crash-risk forecasting difficult, and crash records are too sparse to train reliable models. We present a pipeline converting smartphone traffic video into frame-level crash-risk labels (*safe, caution, risk*) via surrogate safety measures (time-to-collision, closing speed), from YOLOv8 detection and ByteTrack tracking. We build per-frame vehicle-interaction graphs and propose an Edge-Aware Graph Attention Network (GAT) with a BiLSTM to jointly learn spatial interactions and temporal evolution. On 77,592 tracked detections and 513,035 interactions from five smartphone videos (5,071 labeled frames, 5,026 graph sequences), the proposed GAT+BiLSTM achieves the best test performance (accuracy 0.821, macro-F1 0.695), ahead of Random Forest (0.756, 0.669), Logistic Regression (macro-F1 0.500), and a graph-free BiLSTM ablation (macro-F1 0.439), isolating a 25.6-point gain from graph attention. Permutation importance and GAT attention heatmaps provide explainability, showing dataset scale was the key constraint enabling the proposed model to overtake classical baselines.

Index Terms—crash risk prediction, graph attention network, BiLSTM, surrogate safety measures, explainable AI, mixed traffic, intelligent transportation systems, Bangladesh

I. INTRODUCTION

Road traffic injuries remain one of the leading causes of death worldwide, and the burden falls disproportionately on low- and middle-income countries with heterogeneous, non-lane-based traffic [1]. Dhaka’s traffic is a representative case: rickshaws, buses, private cars, motorcycles, and pedestrians share the same carriageway with little lane discipline, producing highly dynamic, close-proximity interactions that traditional, crash-log-based prediction models cannot capture. Two structural problems compound the issue. First, historical crash records in Bangladesh are sparse, inconsistently reported, and rarely geo-referenced at the intersection level, which rules out the large labeled crash datasets that data-hungry deep models typically rely on. Second, most existing crash-risk systems are *reactive*: they detect a collision after it has already occurred, rather than forecasting risk far enough in advance to allow a warning or intervention [2].

This motivates a proactive, video-based alternative: instead of learning from rare crash events, we derive dense, physically interpretable risk labels directly from everyday traffic video using surrogate safety measures (SSMs) such as time-to-collision (TTC) and distance at closest point of approach (DCPA) [3],

and use these labels to train models that forecast near-miss risk several seconds ahead. Building on our project proposal, we implement a full pipeline: YOLOv8 object detection, ByteTrack multi-object tracking, trajectory and motion feature extraction, pairwise SSM computation, automatic risk labeling, dynamic graph construction, and an Edge-Aware Graph Attention Network combined with a BiLSTM (GAT+BiLSTM) for joint spatial-temporal risk classification, with explainability provided through GAT attention weights and permutation feature importance.

Because our proposed GAT+BiLSTM model is a graph-sequence deep network, it is data-hungry by nature. To rigorously test whether it delivers an advantage over simpler alternatives under the data volume realistically available from a small number of self-recorded videos, we benchmark it against a sequence-only BiLSTM (no graph structure) and two classical machine learning baselines trained on hand-engineered, frame-level aggregate statistics of the same interaction features, all under an identical, leakage-free 70/15/15 split. This benchmarking design – rather than reporting only the proposed model in isolation – is itself a contribution, since it exposes exactly how much (or how little) the graph-attention and temporal-recurrence components add once the same information is available to a much simpler, less data-hungry classifier.

The contributions of this paper are:

- 1) An end-to-end, annotation-free pipeline that converts raw, smartphone-recorded traffic video into frame-level and interaction-level crash-risk labels using YOLOv8 detection, ByteTrack tracking, and TTC/DCPA/closing-speed-based surrogate safety measures.
- 2) An Edge-Aware Graph Attention Network that fuses node (per-object motion) and edge (pairwise interaction) features to model asymmetric vehicle-to-vehicle interactions, combined with a BiLSTM to capture how risk evolves over a 10-frame temporal window.
- 3) A controlled, apples-to-apples comparison of the proposed GAT+BiLSTM model against a BiLSTM-only ablation and classical ML baselines (Logistic Regression, Random Forest) on identical splits, isolating the specific contribution of the graph-attention layers and tracking how the model ranking shifts as the labeled video corpus

grows from four to five videos (2,654 to 5,071 labeled frames).

- 4) An explainability analysis combining permutation feature importance (for the classical baseline) and GAT attention heatmaps (for the proposed model), showing that minimum time-to-collision is the single most informative risk signal in both cases.

II. RELATED WORK

Several recent works target spatiotemporal crash-risk or crash-detection modeling. Erkartal and Yilmaz [2] propose a hybrid deep-learning/probabilistic framework (R-CNN, Kalman filtering, HMM, ANN) that detects crashes from monocular traffic video with roughly 91% accuracy; however, the system is reactive – it detects a crash *after* impact rather than forecasting risk beforehand – and was validated on a small highway-traffic dataset of 67 videos, not on mixed, non-lane-based traffic. Nippani et al. [4] show that graph neural networks, particularly GraphSAGE, are effective for road-safety modeling (about 87% AUROC) on a large-scale, 9-million-record crash dataset spanning eight U.S. states, and demonstrate that road connectivity is a critical predictor; their formulation, however, is a *static*, road-network-level graph built from historical crash counts, aggregated at a monthly resolution, rather than a dynamic, vehicle-to-vehicle interaction graph suited to second-by-second risk forecasting. Alnowaiser [5] combines CNN, LSTM and GNN components with SHAP-based explainability on a very large (379K-vehicle, 2.5 TB) trajectory dataset and reports strong accuracy (94.5%) and recall (93.7%), but the model is explicitly noted by the author as a black box requiring post-hoc SHAP explanation, depends on costly radar/sensor infrastructure, and is evaluated on structured, lane-based traffic. Li and Chen [6] similarly use deep spatiotemporal trajectory features for accident-risk prediction but likewise target structured road networks with abundant labeled accident data.

Across these works, three gaps stand out for a Bangladesh-relevant setting: (i) most systems are reactive (detect after the fact) rather than proactive (forecast ahead of time); (ii) most assume either abundant historical crash records or costly sensor infrastructure, both of which are unavailable in our context; and (iii) none directly address unstructured, non-lane-based mixed traffic, where vehicle size and behavior asymmetry (e.g., a bus versus a rickshaw) is the norm rather than the exception. We address the first two gaps by deriving proactive, frame-level risk labels purely from ordinary smartphone video via surrogate safety measures, and the third by using an edge-aware Graph Attention Network [7], whose learned attention weights are naturally suited to asymmetric road-user interactions, rather than a fixed-topology GNN. Our detection and tracking stage builds on the YOLOv8 detector [8] and the ByteTrack multi-object tracking algorithm [9], which in turn builds on the association-metric tracking approach of Wojke et al. [10]. For explainability we use permutation feature importance [11] for the classical baseline and raw GAT attention weights, in the spirit of gradient- and activation-

based explanation methods such as Grad-CAM [12], for the proposed spatiotemporal model. Unlike the works above, we explicitly benchmark the proposed graph-based deep model against strong classical baselines rather than reporting it in isolation, and we track how its ranking shifts as the labeled video corpus grows – which we believe is a more useful and reproducible contribution for practitioners deciding how much self-collected video data is “enough” before a graph-attention model is worth deploying.

III. DATASET AND DATA COLLECTION

A. Sources

We use two data sources. The *Bangladeshi Traffic Flow Dataset* (Mendeley Data, via its Kaggle mirror; 47,356 raw traffic images from multiple Dhaka locations) was used only to sanity-check a pretrained YOLOv8n detector on local, heterogeneous traffic imagery – it was *not* used to fine-tune the detector and contributes no risk labels or trajectories.

All trajectory, interaction, and risk-label data instead come from **five traffic videos recorded on ordinary smartphones from pedestrian overbridges** at different locations in and around Dhaka – not fixed CCTV. All five, including one long airport-corridor video (`Airport.MOV`, 2,417 processed frames, excluded from an earlier iteration purely for processing-time reasons), are now processed end-to-end (Table I).

TABLE I: Smartphone traffic videos processed by the pipeline.

Video	Processed frames	Tracked detections
Airport	2,417	33,712
Rajlokkhi	1,219	22,207
Ajampur1	1,208	20,344
Dhaka test clip	102	1,084
Arambag (double lane)	203	245
Total	5,149	77,592

Frames were sampled with stride 3. After class filtering (Section III-B), 77,592 valid detections across 2,102 unique tracked objects remained, spanning six road-user classes. Adding the airport-corridor video roughly doubled the dataset and raised scene density (15–20 simultaneously visible road users per frame vs. 3–8 elsewhere), widening the range of interaction complexity available for training.

B. Preprocessing and Feature Extraction

Each video is processed frame-by-frame with a pretrained YOLOv8n detector [8] coupled to ByteTrack [9] (confidence threshold 0.35, persistent identity association). We record frame index, timestamp, object identity, class, and bounding box for every detection. Spurious COCO classes (e.g., *boat*, *bench*) are filtered, retaining six road-user classes and reducing 77,754 raw detections to 77,592 valid rows. For every tracked object we compute displacement, speed, acceleration (3-frame rolling mean), and velocity components. For every co-occurring object pair per frame we compute distance, relative/closing speed, time-to-closest-approach (TCA), time-to-collision (TTC), distance at closest approach (DCPA), and a size-based safety radius – yielding **513,035 interaction samples**.

Automatic risk labeling: We derive interaction-level risk labels from the SSMs using a 5-second horizon H : *risk* if

$$TCA \leq 2.0 \text{ s} \wedge DCPA \leq 1.0 r_{\text{safety}} \wedge v_{\text{closing}} > 0, \quad (1)$$

caution if within horizon and moderately close, or already close:

$$(TCA \leq H \wedge DCPA \leq 1.5 r_{\text{safety}} \wedge v_{\text{closing}} > 0) \vee (d \leq 1.25 r_{\text{safety}}), \quad (2)$$

and *safe* otherwise. A continuous score $s = 0.45s_{\text{time}} + 0.40s_{\text{dist}} + 0.15s_{\text{speed}} \in [0, 1]$ is also computed. This gave 359,124 safe, 102,607 caution, 51,304 risk interaction labels. Interaction labels aggregate to a **frame-level** label (max risk rank per frame, plus summary statistics), giving **5,071 labeled frames**: 3,549 safe, 1,014 caution, 508 risk (7:2:1 imbalance preserved across the stratified split: train 3,549, val 761, test 761).

C. Dynamic Graph Construction

Each frame is a graph with up to $M=20$ nodes (padded/masked). Nodes carry a 9-dim feature vector (position, size, class, velocity, speed, acceleration); edges carry the pairwise SSM features above. Frames group into overlapping $T=10$ -frame sequences per video, giving 5,026 graph sequences, split 70/15/15 into training (3,518), validation (754) and test (754).

IV. METHODOLOGY

A. Overview

Fig. 1 shows the overall pipeline: video $\rightarrow$ detections $\rightarrow$ trajectories $\rightarrow$ interaction features/risk labels $\rightarrow$ dynamic per-frame graph. We evaluate (i) classical ML on hand-engineered frame-level aggregate statistics, and (ii) the proposed edge-aware GAT+BiLSTM on the graph sequences directly, with explainability via permutation importance and GAT attention weights respectively.

B. Classical Machine Learning Baselines

For each labeled frame we compute 27 hand-engineered aggregate features: edge count, distance/TCA/TTC/DCPA statistics (min/mean, plus DCPA-to-safety-radius ratio), closing/relative-speed statistics, object count, class diversity, per-class counts, and motion statistics. Two classifiers are trained under the same 70/15/15 split: **Logistic Regression (LR)**, a linear, class-balanced baseline (up to 1,000 iterations); and **Random Forest (RF)** [13], 300 class-balanced trees of max depth 10, chosen as a strong non-linear tabular baseline robust to threshold-like TTC/DCPA risk. Neither model sees the interaction-level risk score/label itself (which would leak the target).

C. Sequence Baseline: BiLSTM

To isolate the graph-attention contribution, we first train a BiLSTM [14] on pooled per-frame node features (masked mean over up to 20 objects, no edge modeling): one bidirectional LSTM layer (64 units/direction) plus a two-layer head, Adam ($\text{lr}=10^{-3}$), dropout 0.3, batch 64, class-weighted cross-entropy, early stopping on validation macro-F1 (max 80 epochs). 57,859 trainable parameters.

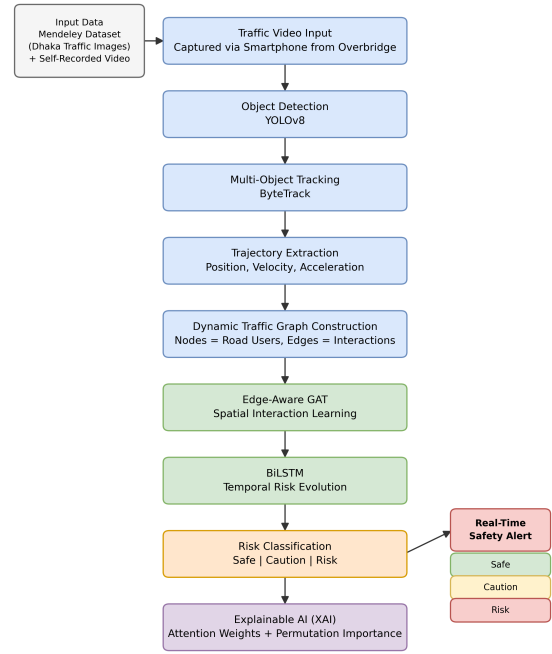


Fig. 1: Overall pipeline: from smartphone traffic video to explainable crash-risk classification.

D. Proposed Model: Edge-Aware GAT + BiLSTM

Each frame's graph passes through two stacked **Edge-Aware GAT** layers before temporal aggregation. For node features h_i and edge features e_{ij} (both projected to 64-dim), the attention score for pair (i, j) is

$$\alpha_{ij} = \text{softmax}_j \left(\text{MLP}([h_i \parallel h_j \parallel e_{ij}]) \right), \quad (3)$$

with a two-layer LeakyReLU MLP and softmax over valid neighbors (including self-loops); nodes update as $h'_i = \text{ELU}(\sum_j \alpha_{ij} h_j)$. After two such layers, masked mean pooling yields one 64-dim graph embedding per frame. The $T=10$ -step embedding sequence feeds a single-layer BiLSTM (64 units/direction), followed by a two-layer head (64 units, ReLU, dropout 0.3) for the 3-way prediction. The model has 106,053 trainable parameters (roughly double the BiLSTM baseline). Training: Adam ($\text{lr}=10^{-3}$), class weights $[0.477, 1.663, 3.313]$, dropout 0.3, batch 64, early stopping on validation macro-F1 (max 80 epochs).

E. Explainability

For Random Forest we compute permutation feature importance [11] (20 repeats, macro-F1 scoring) on the test set. For GAT+BiLSTM we extract the raw attention matrix α_{ij} from the final GAT layer, visualized as a heatmap over source/target object pairs.

F. Evaluation Protocol

All models use the same stratified 70/15/15 split. We report accuracy, macro-averaged precision/recall/F1 (given the 7:2:1 imbalance), weighted-F1, and macro-averaged ROC-AUC (one-vs-rest).

V. RESULTS AND ANALYSIS

A. Overall Comparison

Table II and Fig. 2 summarize test-set performance on the expanded, five-video dataset. The proposed **GAT+BiLSTM is now the strongest model overall**, reaching 82.1% accuracy and 0.695 macro-F1, narrowly ahead of Random Forest (75.6%, 0.669) and clearly ahead of Logistic Regression (macro-F1 0.500) and the sequence-only BiLSTM (macro-F1 0.439). Random Forest retains a marginally higher macro ROC-AUC (0.890 vs. 0.889), so the top two are effectively tied on ranking ability, but GAT+BiLSTM produces better-calibrated hard decisions. This reverses the ordering on the earlier, four-video/2,654-frame dataset, where Random Forest led by a wide margin (macro-F1 0.748 vs. 0.458) – analyzed in Section V-B.

The controlled ablation against the BiLSTM-only baseline is the cleanest evidence here: both models share an identical temporal head, training procedure, and data, differing only in whether per-frame features pass through two edge-aware GAT layers or simple masked mean pooling. The gap is now much larger than before – macro-F1 0.695 vs. 0.439, a 25.6-point absolute (58% relative) improvement, up from 5.1 points on the smaller dataset – showing the graph-attention mechanism was starved for data earlier and now extracts substantially more signal than uniform pooling.

TABLE II: Test-set performance, all models (70/15/15 split, 5-video dataset).

Rank	Model	Acc.	Macro F1	Wtd. F1	Macro AUC
1	GAT + BiLSTM (proposed)	0.821	0.695	0.812	0.889
2	Random Forest	0.756	0.669	0.767	0.890
3	Logistic Regression	0.615	0.500	0.642	0.770
4	BiLSTM (sequence only)	0.606	0.439	0.620	0.696

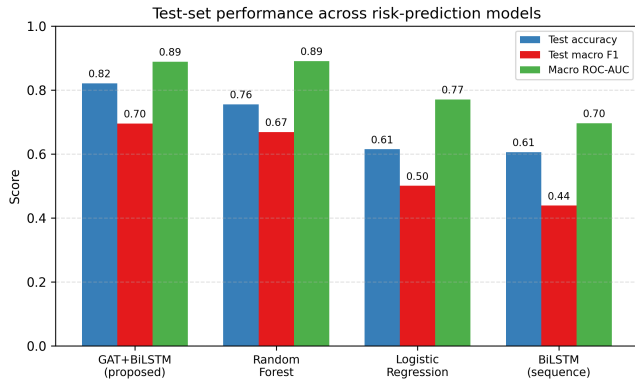


Fig. 2: Test accuracy, macro-F1 and macro ROC-AUC, all models, ranked by macro-F1.

Table III breaks down class-wise metrics (from the confusion matrices in Fig. 3). GAT+BiLSTM is strongest on *safe* (F1 0.911, recall 94.7%) and *caution* (F1 0.572), overtaking Random Forest on both. On the safety-critical *risk* class it has the highest precision (78.7%, few false alarms) but lower recall than Random Forest (48.7% vs. 67.1%): Random Forest catches more true risk frames but with more false positives (63.8% precision). These are complementary strengths

– GAT+BiLSTM is the more trustworthy alarm once it fires, Random Forest the more sensitive detector. BiLSTM-only remains weakest on every class, confirming the graph-attention layers, not just the recurrence head, drive GAT+BiLSTM’s gains.

TABLE III: Class-wise test-set metrics.

Model	Class	P	R	F1
GAT + BiLSTM	Safe	0.879	0.947	0.911
	Caution	0.597	0.550	0.572
	Risk	0.787	0.487	0.602
Random Forest	Safe	0.903	0.824	0.862
	Caution	0.436	0.559	0.490
	Risk	0.638	0.671	0.654
BiLSTM (seq. only)	Safe	0.795	0.734	0.763
	Caution	0.299	0.311	0.305
	Risk	0.209	0.303	0.247

B. Why GAT+BiLSTM Overtakes Random Forest at This Data Scale

Why does the ranking reverse relative to the smaller, four-video dataset (Random Forest 0.748 vs. GAT+BiLSTM 0.458)? Two factors stand out.

Dataset size relative to model capacity: The labeled set nearly doubled (2,654→5,071 frames; 266→508 risk frames). With 106,053 trainable parameters, the earlier data scale gave roughly 40 parameters per risk-labeled example – a regime in which a randomly initialized attention mechanism struggles. At the current scale this ratio roughly halves, matching the sharper, better-calibrated attention pattern in Fig. 4(b) versus the earlier, diffuse pattern. This confirms our earlier hypothesis that the graph model was primarily data-starved, not architecturally unsuited – adding the excluded video was the highest-leverage fix.

Scene complexity favors explicit interaction modeling:

The airport video is markedly denser (15–20 visible road users/frame vs. 3–8 elsewhere). Random Forest consumes this only through fixed, pre-aggregated statistics (e.g., *minimum* TTC across all pairs), discarding *which* pair is interacting. GAT computes attention over all pairwise relationships and can weight specific interactions by context; Fig. 4(b) shows attention now sharply concentrated on a few source-target pairs rather than diffuse – a capability fixed aggregation structurally lacks. Fig. 4(a) confirms *min_ttc_sec* remains Random Forest’s dominant feature, and (c) shows its per-class ROC-AUC is still strong (0.907/0.814/0.950) – ranking ability held steady even as accuracy and macro-F1 *fell* slightly from the smaller dataset (0.807→0.756, 0.748→0.669), since a single pre-aggregated statistic increasingly conflates different situations as scene density rises.

C. Summary

Ranking by macro-F1: GAT+BiLSTM (0.695) > RF (0.669) > LR (0.500) > BiLSTM-only (0.439). GAT+BiLSTM is now the strongest model once trained on a sufficiently large corpus and should be the preferred deployment candidate; dataset scale, not architecture, was the binding constraint previously – one added video, no model changes, converted a clear

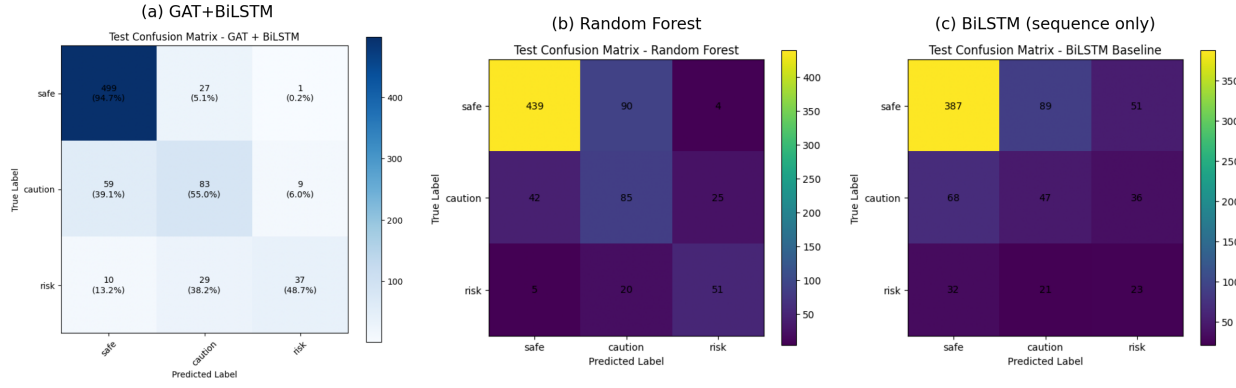


Fig. 3: Test confusion matrices: (a) proposed GAT+BiLSTM, (b) Random Forest, (c) sequence-only BiLSTM baseline.

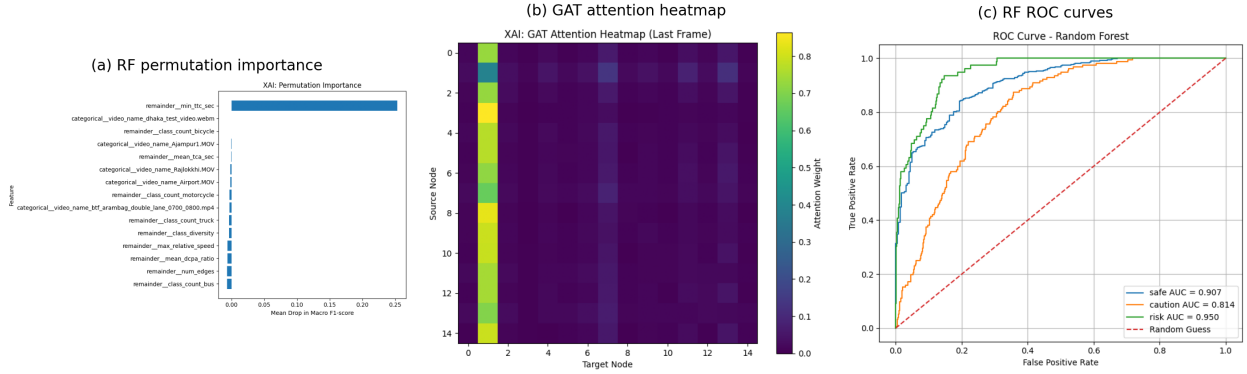


Fig. 4: Explainability: (a) Random Forest permutation importance, (b) GAT attention heatmap (now sharply concentrated), (c) Random Forest per-class ROC curves.

deficit into a clear lead; and Random Forest remains a strong, data-efficient, high-ROC-AUC baseline worth retaining as a fallback or ensemble component.

D. Effect of Dataset Scale: A Direct Comparison

Table IV makes the data-scale argument concrete by placing both pipeline iterations side by side. Nearly every classical-baseline metric is stable or slightly down at the larger scale, while every GAT+BiLSTM metric improves sharply – exactly the asymmetry expected if the graph-attention model, but not the tree-based classifier, was capacity-constrained by data volume rather than by architecture.

TABLE IV: Four-video vs. five-video dataset: scale and resulting performance.

Quantity	4-video iteration	5-video iteration
Labeled frames	2,654	5,071
Risk-labeled frames	266	508
Graph sequences	2,618	5,026
Params per risk frame (GAT+BiLSTM)	~399	~209
Random Forest macro-F1	0.748	0.669
GAT+BiLSTM macro-F1	0.458	0.695
BiLSTM-only macro-F1	0.469	0.439
Best model	Random Forest	GAT+BiLSTM

E. Limitations and Threats to Validity

Several limitations should temper the conclusions above. First, all risk labels are generated automatically from surrogate safety measures rather than verified against real crash

or near-miss records; TTC/DCPA-based labeling is a well-established proxy [3], but it has not been validated against ground-truth incidents on these specific corridors, so absolute performance numbers should be read as relative comparisons between models rather than as calibrated real-world risk probabilities. Second, despite nearly doubling, the dataset still comes from only five videos at five physical locations; the train/validation/test split is stratified by label but not by video, so some scene-specific visual patterns could leak across splits, and performance on genuinely unseen locations may be lower than reported here. Third, the automatic labeling rule and the frame-level aggregate features given to the classical baselines share the same underlying TTC/DCPA computation, which is by design (Section V-B) but means the classical baselines benefit from a form of feature-label alignment that a deployed system without ground-truth SSMs would not have. Fourth, all recordings were taken during daytime in dry conditions from a fixed overbridge vantage point; performance under night, rain, or fog, or from a moving-camera (e.g., dashcam) viewpoint, remains untested. Fifth, we report single-run results per model rather than averages over multiple random seeds, so the exact magnitude of the reported gaps (though not their direction, which is corroborated by the internally consistent confusion-matrix recomputation in Section V-B) carries some run-to-run variance typical of small-data deep learning. We view

these as natural next steps rather than flaws that undermine the comparison, which remains internally controlled and fair across all four models.

VI. CONCLUSION AND FUTURE WORK

We presented a complete pipeline for proactive, video-based crash-risk forecasting in mixed traffic – YOLOv8 detection, ByteTrack tracking, SSM-based automatic risk labeling, and an Edge-Aware GAT+BiLSTM classifier – benchmarked against classical ML and a sequence-only BiLSTM under identical, leakage-free splits. On an expanded five-video, 5,071-frame dataset, GAT+BiLSTM achieved the best overall performance (macro-F1 0.695, accuracy 0.821), reversing an earlier result on a smaller four-video dataset where Random Forest had led by a wide margin and the proposed model trailed every baseline. The reversal came purely from adding one denser video, with no architectural changes, confirming our earlier diagnosis that the graph model was data-starved rather than poorly suited to the problem. Attention heatmaps are now visibly sharper, and permutation importance still shows minimum TTC as Random Forest’s dominant signal, though Random Forest’s accuracy and macro-F1 slipped slightly on the denser scene distribution even as its ROC-AUC held steady.

The model is not yet deployment-ready: *risk*-class recall (48.7%) still trails Random Forest’s (67.1%). Priorities going forward: (i) continue adding video from more locations, times, and weather, since data scale was the dominant factor here; (ii) combine both models’ complementary strengths in an ensemble or two-stage cascade (Random Forest as a high-recall filter, GAT+BiLSTM as a high-precision, explainable confirmatory stage); (iii) further ablations (GAT layer/head count, auxiliary SSM regression) to characterize returns to scale and close the risk-recall gap architecturally; and (iv) Grad-CAM-style overlays on the source video [12], combined with the sharper GAT attention weights, to highlight *which vehicles* triggered an alert directly on the frame – closing the loop toward a real-time, explainable road-safety tool for Dhaka’s mixed-traffic corridors.

REFERENCES

- [1] World Health Organization, “Global status report on road safety 2023,” *World Health Organization*, 2023.
- [2] R. B. Erkartal and A. Yilmaz, “A novel hybrid deep learning-probabilistic framework for real-time crash detection from monocular traffic video,” *Applied Sciences*, vol. 15, p. 10523, 2025.
- [3] M. M. Minderhoud and P. H. L. Bovy, “Extended time-to-collision measures for road traffic safety assessment,” *Accident Analysis & Prevention*, vol. 33, no. 1, pp. 89–97, 2001.
- [4] A. Nippani, D. Li, H. Ju, H. Koutsopoulos, and H. Zhang, “Graph neural networks for road safety modeling: Datasets and evaluations for accident analysis,” in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.
- [5] K. Alnowaiser, “A computational intelligence gnn-lstm framework for spatiotemporal prediction of traffic accident severity in smart cities using shap xai,” *International Journal of Computational Intelligence Systems*, vol. 18, no. 271, 2025.
- [6] H. Li and L. Chen, “Traffic accident risk prediction based on deep learning and spatiotemporal features of vehicle trajectories,” *PLOS ONE*, vol. 20, no. 5, p. e0320656, 2025.

- [7] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, “Graph attention networks,” in *International Conference on Learning Representations (ICLR)*, 2018.
- [8] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics yolov8,” <https://github.com/ultralytics/ultralytics>, 2023.
- [9] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, “Bytetrack: Multi-object tracking by associating every detection box,” in *European Conference on Computer Vision (ECCV)*. Springer, 2022, pp. 1–21.
- [10] N. Wojke, A. Bewley, and D. Paulus, “Simple online and realtime tracking with a deep association metric,” in *2017 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2017, pp. 3645–3649.
- [11] A. Fisher, C. Rudin, and F. Dominici, “Random forests and permutation importance,” *Journal of Machine Learning Research*, vol. 20, no. 177, pp. 1–81, 2019.
- [12] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-cam: Visual explanations from deep networks via gradient-based localization,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.
- [13] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [14] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.